# ZhuLong: An Execution-Grounded LLM Agent for EDA Scripting

Anonymous Author(s)

## Abstract

EDA scripting with tool-specific, often undocumented APIs remains a long-tail bottleneck. We present **ZhuLong**, an execution-grounded LLM coding agent for PyAether, SKILL, and Tcl that combines API retrieval, documentation inspection, and sandbox execution via unified MCP tools into a closed generate–execute–refine loop. We formalize the "grounding gap": EDA correctness is a *situated* property that cannot be guaranteed from static knowledge, bounded by observation fidelity—not model scale.

On **EDA-Eval-PyAether** (158 real-world tasks), the complete system achieves **81.4%** Pass@1 (±1.5), outperforming a pure LLM baseline (**10.5%**). Ablation shows the execution loop is decisive: adding sandbox execution to retrieval yields +9.2 pp, while removing it (−9.2 pp) or retrieval (−0.4 pp) reduces accuracy—consistent with the theoretical prediction. A cross-language reproduction on the SKILL and Tcl slices of **EDA-Scripting-Bench** (449 tasks total) reproduces the same structure—the complete system is the top and the prior-only baseline the bottom on every language.

## CCS Concepts

• **Hardware** → **Software tools for EDA**.

## Keywords

LLM Agents for EDA, EDA Scripting, Code Generation, Sandbox Execution, Execution Grounding

**ACM Reference Format:**
Anonymous Author(s). 2026. ZhuLong: An Execution-Grounded LLM Agent for EDA Scripting. In *Proceedings of* . ACM, New York, NY, USA, 8 pages.

## 1 Introduction

Large language models (LLMs) have demonstrated strong capabilities in general-purpose code generation [17, 23, 25]. However, applying LLMs to EDA scripting—where engineers routinely manipulate in-memory design databases, create layout instances, route schematics, and query netlists—remains challenging. First, EDA APIs are long-tailed and tool-specific; many rarely appear in public training corpora. Second, correctness cannot be verified statically: it depends on observable side effects in a real EDA environment, such as modified layouts or updated schematics. Third, without execution feedback, static generation cannot diagnose errors or iteratively refine incorrect code [3, 5]. Existing LLM-based EDA work has mainly focused on HDL generation [15, 21], while scripting with real tool APIs—especially with incomplete documentation and sandbox execution—remains underexplored.

This paper presents **ZhuLong**, an execution-grounded LLM coding agent for PyAether, SKILL, and Tcl. Named after a mythical creature whose opened eyes bring light, ZhuLong aims to illuminate opaque EDA APIs and automate tool-specific scripting workflows. Built on Cline-CLI [7], ZhuLong extends three MCP tools [16]—`search_apis`, `get_api_details`, and `run_code`—that form a closed loop: retrieve candidate APIs, inspect their documentation, execute code in the EDA environment with execution feedback, and iteratively refine based on observed errors or outputs. All tools accept a unified `lang` parameter, so the same agent logic covers PyAether, SKILL, and Tcl. Beyond the engineering, we provide a **theoretical foundation** (Section 4): EDA scripting correctness is a *situated* property that cannot be guaranteed from static knowledge, and a Bayesian analysis shows the resulting "grounding gap" is bounded by observation fidelity—not model scale—explaining why runtime observation is essential, and why the Harness must supply both halves of it: reading the environment's documented API facts, and reading back what executing code does to the live design.

The main contributions are:

- **An account of *which* factor sets the ceiling, with measurements that distinguish them (RQ1).** We formalize the *grounding gap*: EDA correctness is a situated (L4) property that no offline knowledge can guarantee, bounded by *observation fidelity* not model capacity (Section 4). We answer RQ1 from both sides—laddering $\Omega$ with the LLM fixed, and changing $\mathcal{K}$ with the harness fixed—stated *before* the measurements as predictions the experiments could refute.
- **A harness whose necessary components we isolate (RQ2).** ZhuLong couples three MCP tools—`search_apis`, `get_api_details`, `run_code`—so that retrieval (observation), sandbox execution (effector), and the generate–execute–refine loop (belief update) form one closed loop, driven by a single `lang` parameter covering PyAether, SKILL, and Tcl (Section 3). ZhuLong attains **81.4%** Pass@1 on the 158-task PyAether slice, versus **10.5%** (pure LLM) and **72.2%** (RAG). Ablation shows the components are jointly necessary and not interchangeable: adding execution to RAG yields **+9.2 pp**, removing it (−9.2 pp) or retrieval (−0.4 pp) reduces accuracy, and bounding $\Phi$ degrades it monotonically.
- **A cross-language benchmark on which the same structure can be re-measured per toolchain (RQ3). EDA-Scripting-Bench** places PyAether, SKILL and Tcl under one task schema, one assertion criterion and one harness (Section 5). The completed PyAether slice (**EDA-Eval-PyAether**, 158 expert-reviewed tasks) is the first standardized PyAether benchmark. We release the harness, task schema and

evaluation protocol; task cases derive from vendor documentation and customer designs and cannot be redistributed.

## 2 Background and Motivation

We target two widely-used EDA scripting interfaces: PyAether for Empyrean Aether [9] and SKILL for Cadence Virtuoso [20]. EDA scripting poses three challenges beyond what generic LLM code generators address: (i) APIs are long-tailed, tool-specific, and poorly documented; (ii) correctness depends on side effects in a live, stateful database, not return values; and (iii) no benchmark exists to evaluate whether generated scripts actually work.

**Prior work stops short of execution.** LayoutCopilot [14], ChaTCL [19], and RAG-EDA [18] combine LLMs with static knowledge bases, but none executes generated code: correctness is assessed by human judgment or retrieval quality, not by whether the script actually modifies a layout. Recent concurrent benchmarks for Tcl/Innovus flows [13, 24] confirm the growing recognition of this gap, yet target different toolchains; a dedicated PyAether/SKILL benchmark remains absent.

**Our approach.** ZhuLong closes the loop from generation to execution to refinement via sandbox feedback, with a **theoretical grounding** (Section 4) explaining *why* execution is necessary, and **EDA-Eval-PyAether**, the first standardized PyAether benchmark. The contribution boundary is the *harness*—the three MCP tools (`search_apis`, `get_api_details`, `run_code`), the evaluation protocol, and the benchmark; the agent backbone is outside the scope of this paper.

## 3 System Design

### 3.1 Overall Architecture

ZhuLong is built on Cline-CLI [7], extended with EDA-specific retrieval, documentation, and execution capabilities. We call the resulting runtime an **ICL-type autonomous agent**: it decides its own next action, keeps no explicit posterior, and maintains its belief state implicitly by in-context learning (formalized in Section 4). As shown in Figure 1, the system comprises three components—the LLM-based agent runtime, the API knowledge base, and the EDA execution environment—interacting through three MCP tools.

### 3.2 Three MCP Tools

ZhuLong extends three MCP tools [1], each accepting a unified `lang` parameter (`pyaether`, `skill`, or `tcl`) to support all three platforms with identical agent logic:

- `search_apis(query, lang)` retrieves relevant APIs via semantic and keyword-based search. We index APIs by name and description using BGE-M3 embeddings [4] with FAISS [8] for efficient similarity search, returning the top-$K$ results ($K = 5$ by default) with confidence scores and descriptions.
- `get_api_details(api_name, lang)` fetches comprehensive documentation for a specified API, including parameter types, return values, and examples.
- `run_code(code, mode, lang)` executes the generated code in the EDA environment. It supports two modes: (1) **lightweight**—runs in a sandboxed environment without a GUI session (standalone PythonE for PyAether, or SKILL in batch mode for Virtuoso), capturing outputs, errors, and state changes for iterative refinement; (2) **interactive**—runs inside an open EDA session via the agent-EDA bridge (socket communication), enabling operations on unsaved layouts and schematics.

### 3.3 Agent Workflow

The agent iteratively refines its output through execution feedback [7]. A typical trajectory reasons over the user instruction, retrieves APIs via `search_apis`, inspects candidates via `get_api_details`, generates code, executes it via `run_code`, observes the result, and re-plans—revising code, retrieving alternatives, or terminating—until success or further attempts are deemed unlikely. The agent is not constrained to a fixed sequence: it may skip retrieval when it already knows the required API, or conduct multiple retrieval rounds when initial results lack confidence.

## 4 Theoretical Grounding: The Grounding Gap

Our empirical results show that the execution loop is decisive: adding sandbox execution to retrieval yields +9.2 pp ($p<10^{-8}$), while removing execution (−9.2 pp) or retrieval (−0.4 pp, n.s.) reduces accuracy. This section explains *why* this is structural rather than implementation-specific. We condense the *grounding gap*: EDA scripting correctness is a *situated* property that cannot be guaranteed from offline knowledge, and execution is the only channel that observes the current design state. The gap is bounded by observation fidelity, not model scale.

### 4.1 A Four-Level Hierarchy of Information

To make precise what can be learned offline versus what must be observed online, we distinguish four levels of information (Table 1):

- **L1–Syntactic Correctness**: well-formedness of symbols per formal rules; acquirable offline.
- **L2–Physical Plausibility**: robust statistical patterns of how the world typically behaves; acquirable offline as the agent's *prior*.
- **L3–Historical Factuality**: what has actually occurred or is documented; past-oriented, acquirable offline, but with no connection to what is true *now*.
- **L4–Situated Truthfulness**: what is true *right now* in this specific environment. Three characteristics: *indexicality* (truth depends on context and changes over time), *non-guaranteeability from offline data*, and *real-time commitment*.

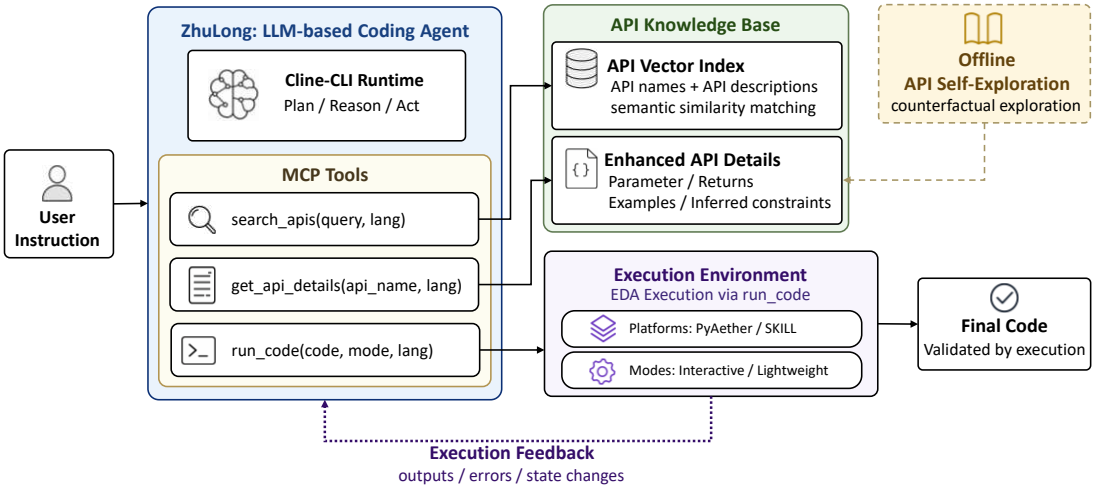



**Figure 1: ZhuLong system architecture. The agent interacts with API retrieval, documentation, and EDA execution via three unified MCP tools. Execution feedback supports iterative refinement.**

Two clarifications: L4 is not a semantic category but a *grounding requirement* imposed by indexical reference (the same proposition may be L3 at one moment, L4 at another); and the four levels separate what can be *known in advance* from what must be *observed now*—L4 is read from the environment, not recalled from memory.

For EDA scripting, the situated state is the live design database: which cell and view are selected, what the layout contains at this moment. The correctness criterion—generated code executes without error and all assertions over the mutated design pass—is an L4 property: a script that is well-formed (L1), plausible (L2), and invokes documented functions (L3) can nonetheless violate an undocumented binding contract or depend on an internal selection state. Such failures must be *observed* by acting on the live design and reading back the result.

## 4.2 A Bayesian Formalization of the Grounding Gap

We cast grounding as decision-making under partial observability. The formulation is diagnostic—it makes precise *what* is missing: the observation likelihood $\Omega$ and the recursive update $\Phi$.

Let $\Theta$ denote the space of L4 states, $o_t$ a runtime observation, $a_t$ a code action, and $\mathcal{K}$ the offline knowledge in the LLM's frozen parameters (L1–L3). The agent maintains a belief

$$b_t(\theta) \triangleq p(\theta_t \mid o_{1:t},\, a_{1:t-1},\, \mathcal{K}), \tag{1}$$

updated by Bayes' rule:

$$b_t(\theta) \propto p(o_t \mid \theta_t) \int_{\Theta} p(\theta_t \mid \theta_{t-1}, a_{t-1})\, b_{t-1}(\theta_{t-1})\, d\theta_{t-1}, \tag{2}$$

where $p(o_t \mid \theta_t)$ is the *observation likelihood* $\Omega$ (the Harness's observation modules) and $p(\theta_t \mid \theta_{t-1}, a_{t-1})$ the *transition model* $\mathcal{T}$ (the LLM's prior over how EDA operations transform the design). Actions minimize expected cost:

$$a_t^* = \arg\min_{a \in \mathcal{A}} \mathbb{E}_{\theta_t \sim b_t}\big[\mathcal{C}(\theta_t, a, G) + \gamma\, V(b_{t+1})\big]. \tag{3}$$

The *grounding gap* is the divergence between the posterior with runtime observations and the posterior from the offline prior alone:

$$D_{\text{grounding}}(t) \triangleq D_{\text{KL}}(p(\cdot \mid o_{1:t},\, \mathcal{K},\, \text{env}) \,\|\, p_{\mathcal{K}}(\cdot \mid o_{1:t}))\,. \tag{4}$$

This divergence is bounded by observation fidelity, not model capacity. If $I(\Theta_t; O_t \mid \mathcal{K}) = 0$—the observation channel is silent—no prior improvement can recover the current state. Scaling $\mathcal{K}$ improves L1–L3 but cannot push the gap below the floor imposed by $\Omega$. The source is causal: parameters are fixed before deployment, and the environment at time $t$ is not a function of the training corpus. This separates grounding from learning-based adaptation: fine-tuning improves the prior over L1–L3 during *training*; grounding requires the likelihood and a posterior update from observing $o_t$ during *deployment*. A better prior does not eliminate the need for observation.

## 4.3 ZhuLong as an Instance of the Agent Harness

The formulation maps onto ZhuLong's architecture: the frozen LLM supplies $\mathcal{K}$ (L1–L3), $\mathcal{T}$, and $\Pi$; the Harness supplies observation, belief update, and execution. `search_apis`/`get_api_details` are the observation modules $\Omega$; `run_code` is the effector acting on the live design database; the re-planning loop implements $\Phi$. The closed loop $b_t \to a_t \to o_{t+1} \to b_{t+1}$ turns one-shot generation into grounded search.

**Table 1: The four levels of information**

| Level | Name | Core Question | Learnable Offline? |
|---|---|---|---|
| L1 | Syntactic Correctness | Is this string well-formed? | Yes |
| L2 | Physical Plausibility | Could this event happen? | Yes |
| L3 | Historical Factuality | Did this event happen? | Yes |
| L4 | Situated Truthfulness | Is this true *right now*? | **No** |

### 4.4 Design Principles

Three principles follow: *freeze* $\mathcal{K}$, *maximize* $\Omega$'s fidelity, and *run* $\Phi$ *continuously*. (1) Decouple knowledge from belief: $\mathcal{K}$ is the static L1–L3 prior in the LLM; $b_t$ is the dynamic posterior maintained by the Harness. Freezing $\mathcal{K}$ and absorbing adaptation in $b_t$ yields stability and flexibility. (2) Optimize observation over the prior: the gap is bounded by $\Omega$'s fidelity, not model scale; observation investment has direct bounded effect, while model scaling has diminishing returns once the prior is rich. (3) Design for continuous belief updating: $\Phi$'s update rate must exceed L4 dynamics, and in an ICL-type autonomous agent the quantity to bound is how much evidence $b_t$ may accumulate—a graded execution budget or a lagged read-back.

These principles *account for* our ablation findings: removing $\Omega$ (w/o Sandbox) makes L4 violations unobservable ($-9.2$ pp); RAG improves the L1–L3 prior but cannot observe L4 (ceiling 72.2%). Three diagnostic questions follow for any candidate system: (1) does it implement $\Omega$ with sufficient fidelity? (2) does it maintain $b_t$ through $\Phi$? (3) does it execute $a_t$ through a causal interface? These organize the evaluation into RQ1–RQ3 (Section 6.1).

## 5 Benchmark Construction

**EDA-Scripting-Bench** spans three EDA scripting languages—PyAether, SKILL, and Tcl—under one task schema and one execution-based protocol. Every task carries a `lang` field and language-appropriate assertions, so a single harness (Section 3.3) can be scored uniformly across platforms. It comprises **EDA-Eval-PyAether** (158 PyAether tasks), a 157-task SKILL slice, and a 134-task Tcl slice (Custom WaveView 77 + Innovus 57), for **449 tasks in total**. Main results are on the PyAether slice; the cross-language reproduction is in Section 6.2.4; the $\Omega$ and $\Phi$ axes are PyAether-only.

**EDA-Eval-PyAether** (the 158-task PyAether slice) was collected from four sources: API references (61.4%), documentation examples (27.2%), internal training materials (7.6%), and anonymized CAD practical cases (3.8%), each manually reviewed by EDA domain experts. Layout and Schematic operations constitute the majority (70.2%: Layout 64 / 40.5%, Schematic 47 / 29.7%; Design Management 24 / 15.2%; Others 23 / 14.6%).

**Evaluation Protocol.** Each task consists of a natural language prompt, a function signature, and assertion-based test code. A task passes if the generated code executes in the PyAether sandbox without errors and all assertions pass. The primary metric is Pass@1 [6]:

$$\text{Pass@1} = \frac{\#\text{ tasks with all assertions passed}}{\text{Total tasks}} \times 100\%$$

## 6 Experiments

### 6.1 Experimental Setup

*6.1.1 Research Questions.* The framework of Section 4 makes three testable claims, one per research question:

- **RQ1 (Ceiling: observation or prior?).** Is accuracy bounded by $\Omega$'s fidelity or by $\mathcal{K}$'s strength? S1 varies $\Omega$ (LLM fixed); S3 varies $\mathcal{K}$ (harness fixed).
- **RQ2 (Which components are necessary, and interchangeable?).** Does bridging the gap require observation, a causal effector, and continuous $\Phi$? S2 varies the effector and the belief-update budget.
- **RQ3 (Domain class or toolchain?).** Does the same structure reproduce on a structurally different toolchain? S1–S2 are re-run on the SKILL and Tcl slices of **EDA-Scripting-Bench**.

RQ1 and RQ2 test the structural claims; RQ3 tests whether they follow from the domain's structure rather than from one toolchain.

*6.1.2 Benchmark and Evaluation Protocol.* We evaluate ZhuLong on **EDA-Eval-PyAether** (158 tasks) using **Pass@1** as the primary metric. Each task runs in a single trace with per-execution timeout 2500 s; the agent may iterate within a trace via $\Phi$ but is never restarted. A file-based `PreToolUse` hook enforces a tool whitelist, blocks shell/code-search, and confines file access to a fresh empty workspace (identical across all arms). Each configuration is evaluated over five independent runs with reshuffled task order; accuracy is mean±std. Generation-stage failures (no executable code) are reported separately from execution-stage failures. Runs with infrastructure faults are discarded and re-run; the full-system anchor must reach $\geq$75% Pass@1 as a health floor. All figures are on the PyAether slice; cross-language results in Section 6.2.4.

*6.1.3 Baselines and ZhuLong Variants.* To isolate each component's contribution, we use two baselines and two ablations (Table 2). Retrieval and documentation form $\Omega$; the sandbox is the effector. Each capability is toggled by a tool-registration switch—a single disabled-list in the server environment—so a disabled tool vanishes from the agent's tool set while every enabled tool keeps a byte-identical implementation across all

arms. The sandbox is removed by unregistering `run_code`; retrieval by unregistering the three retrieval tools; the anti-leak hook stays loaded in every arm.

**Table 2: Configurations and their components**

| Configuration | Retrieval | Sandbox |
|---|---|---|
| Baseline 1 (Pure LLM) | ✗ | ✗ |
| Baseline 2 (RAG) | ✓ | ✗ |
| ZhuLong w/o Retrieval | ✗ | ✓ |
| **ZhuLong** (Full) | ✓ | ✓ |

*6.1.4 LLM Backbones.* All main experiments use DeepSeek-V4-Pro as the primary backbone. To understand the impact of different LLMs, we additionally evaluate GLM-5.2, DeepSeek-V4-Flash, Kimi-K2.6, and Doubao-Seed-2.0-Pro.

## 6.2 Results and Ablations

The framework of Section 4 fixes what each study must vary. Table 3 gives the progressive build-up and the component ablation as an overview; the studies below then change one component at a time, with the LLM frozen unless stated otherwise.

**Table 3: Ablation study results**

| Configuration | Pass@1 (%) | Δ |
|---|---|---|
| *Progressive build-up (vs. previous row)* | | |
| Baseline 1: Pure LLM | 10.5 ± 1.9 | — |
| Baseline 2: RAG | 72.2 ± 1.6 | +61.7 pp |
| **ZhuLong** (Full) | **81.4 ± 1.5** | **+9.2 pp** |
| *Component ablation (vs. ZhuLong Full)* | | |
| ZhuLong w/o Sandbox | 72.2 ± 1.6 | −9.2 pp$^{***}$ |
| ZhuLong w/o Self-Expl. | 77.3 ± 2.9 | −4.1 pp$^{*}$ |
| ZhuLong w/o Retrieval | 81.0 ± 4.5 | −0.4 pp$^{\text{ns}}$ |

Significance from McNemar test on pooled per-task pass/fail across 5 runs (790 paired observations per arm) and paired bootstrap (10 000 resamples, 95% percentile CI): $^{***}$p¡0.001, $^{**}$p¡0.01, $^{*}$p¡0.05, ns = not significant. w/o Sandbox: $p<10^{-8}$, CI $[-13.1, -5.4]$ pp. w/o Self-Expl.: $p<0.05$, CI $[-7.0, -1.2]$ pp. w/o Retrieval: $p=0.81$, CI $[-3.5, +2.9]$ pp (not significant).

*6.2.1 S1: Observation fidelity (RQ1).* Principle 2 predicts that with the model fixed, accuracy is bounded by the fidelity of the observation channel Ω. We vary Ω along the two axes through which it receives information, holding the LLM fixed in both.

**Documentation axis.** Table 4 holds the agent loop and the LLM fixed, and changes the documentation channel together with the presence of the effector: no observation at all; a low-fidelity index built from API names alone, coupled with the full execution loop; and a high-fidelity index that adds descriptions, BM25 term matching, abbreviation expansion and synonym redirect, used without an effector as the RAG baseline.

**Table 4: Documentation fidelity with and without the effector (LLM and loop fixed; (L) includes the effector, (H) is the retrieval-only RAG baseline without it)**

| Rung | Configuration | Pass@1 (%) |
|---|---|---|
| (N) No observation | Baseline 1 (Pure LLM) | 10.5 ± 1.9 |
| (L) Low fidelity | name-only index | 82.8 ± 1.0 |
| (H) High fidelity | + description, BM25, abbreviations, synonyms | 72.2 ± 1.6 |

The documentation axis is non-monotone: (L) at 82.8% exceeds (H) at 72.2%. This reflects the effector, not the index—(L) couples a name-only index with the full execution loop, whereas (H) is retrieval-only without execution. A coarse index with execution outperforms a rich index used only for static retrieval.

**Read-back axis.** We compare three closed-loop variants differing only in how much live state execution returns: (N′) *no observation*—prior only; (B) *binary feedback*—`run_code` reports only pass/fail; (F) *full observation*—stdout, exception traces, per-assertion outcomes, and queried state (the default). The three rungs compress the *same* executed state to different bandwidths: (F) returns the full `error_log`/`print_log`/`result`; (B) collapses it to one bit (`PASS` = no exception, not that assertions held); (N′) returns only `executed`. Principle 2 predicts a monotone gain (N′)<(B)<(F).

Table 5 reports Pass@1 together with the convergence cost (mean `run_code` calls, wall-clock, timed-out tasks) per variant.

**Table 5: Read-back axis: accuracy and convergence cost (LLM and effector fixed). (N′) cost is not recoverable (traces discarded in a storage migration); (F)/(B) cost columns are 5-run means ± sample std.**

| Read-back | Pass@1 (%) | Mean `run_code` | Wall (h) | Time |
|---|---|---|---|---|
| (N′) None (prior only) | 73.3 ± 2.2 | not recoverable$^{\dagger}$ | | |
| (B) Binary pass/fail | 78.1 ± 1.3 | 22.3 ± 3.3 | 3.73 ± 0.72 | 7.6 ± |
| (F) Full observation | 81.4 ± 1.5 | 12.1 | 2.22 | 3.4 |

$^{*}$ mean of 5 runs (range 1–6). $^{\dagger}$ (N′) traces discarded; cost not imputed.

The results recover the predicted monotone ordering (N′)<(B)<(F): N′ → B (+4.8 pp) is the gain from learning whether a run passes at all; B → F (+3.3 pp) is the further gain from seeing the concrete error.[1] Convergence cost moves oppositely: dropping (F)→(B) inflates mean `run_code` calls ∼1.8× (12.1→22.3), wall-clock ∼1.7× (2.22→3.73 h), and timed-out tasks ∼2.2× (3.4→7.6)—the "fidelity down → convergence cost up" trade-off Principle 2 implies.

[1] An earlier legacy probe of the (B) arm showed an N′-over-B inversion (66.2±16.1%, one anomalous run at 43.0%); the fresh 5-run re-measurement collapses the spread to ±1.3 and restores the predicted ordering.

*6.2.2 S2: The closed loop (RQ2).* Principle 3 requires actions to reach the design through a causal interface, and $\Phi$ to run continuously.

**Effector.** Removing the sandbox drops Pass@1 from 81.4% to 72.2% ($-9.2$ pp; McNemar $p<10^{-8}$, 95% CI $[-13.1, -5.4]$ pp); adding it to the RAG baseline raises it by the same amount. Without an effector, $b_t$ receives no evidence about the live state and the agent is reduced to a prior-only predictor. Execution returns concrete error messages that guide corrections, transforming one-shot generation into iterative search [10, 12].

**Belief update.** ZhuLong is autonomous, so we do not shorten the loop or force early submission. Instead we bound how often the loop *may execute*—the number of `run_code` calls per trace—which determines how much state evidence $b_t$ receives. Once the budget is spent, the call is denied at the tool boundary; the agent keeps running under its own control until it submits. Table 6 also includes a *lagged* channel: a single-slot ($L$=1) buffer whose read-back returns the state from the *previous* execution, leaving the agent permanently one action behind the design. Principle 3 predicts accuracy rises with the execution budget but saturates once updates outpace the design's dynamics.

**Table 6: Effect of bounding the evidence available to $\Phi$ (LLM, $\Omega$ and effector fixed). $k$ = max `run_code` calls per trace; "lagged" = read-back returns the *previous* execution's state. "Converged" = share of traces where the agent self-terminated.**

| Update budget | Pass@1 (%) | Converged (%) | Mean gen-fail |
|---|---|---|---|
| $\Phi$ $k$=1 | 63.8 ± 3.2$^{***}$ | 76.3 | 34.6 |
| $\Phi$ $k$=3 | 69.1 ± 2.5$^{***}$ | 80.4 | 27.6 |
| $\Phi$ $k$=10 | 77.7 ± 2.2$^{**}$ | 90.4 | 17.4 |
| $\Phi$ lagged | 73.0 ± 3.2$^{***}$ | 97.5 | 9.0 |
| $\Phi$ unbounded (default) | 81.4 ± 1.5 | 99.1 | 10.4 |

5-run means (Pass@1 ± std, $n$=5); unbounded anchor = full-system baseline (Table 3). *Mean gen-fail* = per-run count of tasks producing no executable code. Significance vs. unbounded (McNemar, paired bootstrap 95% CI): $^{***}$p¡0.001, $^{**}$p¡0.01.

Reading Table 6 with the *Mean gen-fail* column reveals the mechanism: as $k$ shrinks, tasks producing no executable code rise (17.4 at $k$=10, 27.6 at $k$=3, 34.6 at $k$=1), driving Pass@1 downward. The budget is exhausted at the tool boundary, but the agent keeps running; with no further `run_code` calls it cannot finish or validate its generation—the $k\downarrow\rightarrow$ gen-fail$\uparrow\rightarrow$ Pass@1$\downarrow$ signature of $\Phi$ being bounded, consistent with Principle 3.

*6.2.3 S3: The prior axis, the contrast arm of RQ1.* Principle 1 separates knowledge from belief: the frozen prior should bound what a harness can achieve without substituting for observation. Table 7 changes only $\mathcal{K}$—the LLM—with the harness fixed. We use DeepSeek-V4-Pro (FP4) as the primary backbone for its cost–capability trade-off; GLM-5.2 attains marginally higher Pass@1 (83.3%) but overlaps within one standard deviation. Read against Principle 2, this spread is what $\mathcal{K}$ can move—smaller than the observation-channel effects in S1–S2, confirming that the prior does not substitute for observation.

**Table 7: Impact of LLM backbone (ZhuLong)**

| LLM Backbone | Pass@1 (%) | Δ |
|---|---|---|
| DeepSeek-V4-Pro | 81.4 ± 1.5 | — |
| GLM-5.2 | 83.3 ± 3.1 | +1.9 pp$^{ns}$ |
| DeepSeek-V4-Flash$^{\dagger}$ | 60.1 ± 5.5 | -21.3 pp$^{***}$ |
| Kimi-K2.6 | 77.0 ± 1.6 | -4.4 pp$^{**}$ |
| Doubao-Seed-2.0-Pro | 63.8 ± 1.4 | -17.6 pp$^{***}$ |

Significance vs. DeepSeek-V4-Pro (McNemar, paired bootstrap 95% CI): $^{***}$p¡0.001, $^{**}$p¡0.01, ns = not significant. Flash$^{\dagger}$: retest 5-run under reduced gateway parallelism ($-p$ 4) and enlarged token budget; the initial 5-run (11.9%, depressed by an infrastructure confound) is superseded.

*6.2.4 RQ3: re-measuring the structure per toolchain.* The three slices of **EDA-Scripting-Bench** share one task schema, one assertion criterion, and one harness, so S1–S2 can be re-run slice by slice with nothing changed but the toolchain. Table 8 reports the component ablation on SKILL (157 tasks) and Tcl (134 tasks, aggregating WaveView and Innovus), alongside the PyAether anchor; the *Overall* column pools all 449 tasks. The comparison of interest is the *within-language* ordering, not cross-table absolute rates.

**Table 8: Cross-language reproduction of the component ablation (RQ3). Pass@1 (%), 5-run mean±std on PyAether (158 tasks) and SKILL (157 tasks); Tcl aggregates the Custom WaveView (77) and Innovus (57) runtimes into a single 134-task column (task-weighted). The Overall column pools all 449 tasks into one task-weighted Pass@1 per configuration (single value, no run-level dispersion). The PyAether column is the main-table anchor (Table 3). The full system is the top configuration on every language; the prior-only baseline is the bottom on every language.**

| Configuration | PyAether | SKILL | Tcl | Overall |
|---|---|---|---|---|
| Pure LLM | 10.5 ± 1.9 | 33.2 ± 2.3 | 10.2 | 18.3 |
| RAG (retrieval only) | 72.2 ± 1.6 | 33.9 ± 10.6 | 84.6 | 62.5 |
| ZhuLong w/o Retrieval | 81.0 ± 4.5 | 39.0 ± 12.6 | 73.2 | 64.0 |
| **ZhuLong (Full)** | **81.4 ± 1.5** | **60.3 ± 6.7** | **87.2** | **75.8** |

The qualitative structure reproduces across all three languages: the full system is the *top* configuration and the prior-only baseline the *bottom* on every language—the dominance of observation-plus-execution is domain-structural, not toolchain-specific (RQ3). The intermediate ordering is less stable: on PyAether the sandbox-only variant beats RAG, whereas on Tcl the reverse holds and on SKILL the two are close, reflecting that different toolchains expose different amounts of usable documentation relative to executable feedback. The two extremes are invariant, and the *Overall*

column reproduces the same ordering at the pooled level. The (Ω) and (Φ) axes are reported on PyAether only.

### 6.3 Error Analysis

To characterize failure modes, we inspected a representative trace from the full-system 5-run set—the run whose Pass@1 (81.6%, 129/158) is closest to the 5-run mean (81.4±1.5%). The 29 failures comprise 17 generation-stage failures (no executable code produced) and 12 execution-stage failures, classified by the L1–L4 hierarchy of Section 4.

The dominant execution-stage failure mode is **L4** (8/12, 67%): situated truthfulness violations that no offline knowledge can prevent—the code is syntactically correct (L1), uses documented APIs with plausible arguments (L2–L3), yet fails because it violates an *undocumented* binding contract, depends on an internal selection state, or assumes a runtime context that does not hold. For example, Task 010 (`add_instance_to_schematic`) executes and passes the agent's own assertions, yet the official harness rejects it: `target_cv must be an opened cellview object`—an L4 property of the live design state, not of the API signature. Tasks 063, 074, 146, 155 follow analogous patterns (hidden type names, attributes absent in the test path, GUI-dependent return values). The remaining execution-stage failures are **L3** (4/12, 33%): valid APIs composed with incorrect logic (Task 052 substitutes a custom validator for the mandated call; Task 058 uses substring matching instead of equality). **L1/L2 = 0%**.

The 17 generation-stage failures are timeout (4), wrong save path (10, saving to `ws/` instead of `ws/pyAether_code/`—a procedural error amenable to harness-level mitigation), and no code produced (3).

The L1–L4 classification makes the grounding-gap framework concrete: 67% of execution-stage failures are L4, invisible until the agent acts on the live design and reads back the result. This is why the execution loop is decisive: the sandbox is the only channel through which the agent can observe hidden type contracts, internal selection states, and runtime-context dependencies. These findings motivate type-aware API documentation and case-level knowledge distillation.

## 7 Discussion

The error analysis (Section 6.3) shows that 67% of execution-stage failures are L4 (situated truthfulness) violations caused by undocumented API contracts and hidden type signatures, motivating type-aware retrieval and case-level knowledge distillation. More broadly, the dominance of sandbox execution (+9.2 pp) over static knowledge elevates execution grounding from an empirical observation to a structural principle: the grounding gap is bounded by observation fidelity, so execution channels—not larger priors—are the binding constraint for domains with long-tailed, under-documented APIs. This paradigm may generalize to other domain-specific scripting tasks with similar characteristics.

## Limitations

Several limitations apply. (1) Performance varies across base LLMs (60.1%–83.3%); the model remains a decisive factor. (2) The interactive bridge is single-threaded. (3) Main results are on the 158-task PyAether slice; the Ω and Φ axes are PyAether-only (cross-language component ablation in Section 6.2.4). (4) Benchmark size limits statistical power for small effects; we report McNemar/bootstrap for main contrasts but treat small gaps (e.g. −0.4 pp, $p$=0.81) as directional. (5) The primary backbone is FP4-quantized; quantization effect is not ablated. (6) Convergence-cost comparison is complete for (F) and (B) only; (N′) traces were discarded.

## 8 Conclusion and Future Work

**ZhuLong** is an execution-grounded LLM coding agent for PyAether, SKILL, and Tcl that closes the generate–execute–refine loop via sandbox feedback. Its theoretical grounding—EDA correctness is a situated (L4) property, and the grounding gap is bounded by observation fidelity, not model scale—explains why runtime observation is necessary. On **EDA-Eval-PyAether** (158 real-world tasks), the complete system achieves 81.4% Pass@1 (±1.5), substantially outperforming static RAG (72.2%); ablation shows execution and retrieval are jointly necessary. This work establishes three foundations: (1) the grounding-gap theory, (2) the execution-grounded agent paradigm generalizable to other long-tail-API domains, and (3) **EDA-Scripting-Bench** (449 tasks across three toolchains) as evaluation infrastructure. The harness, task schema and protocol will be released.

Future work proceeds along three directions. First, extending all ablation axes to the full EDA-Scripting-Bench (PyAether + SKILL + Tcl, 449 tasks), together with extending the benchmark to systematically cover GUI-bridged tasks on open design sessions. Second, evolving the agent through self-improvement—learning from both successful and failed execution traces to refine its retrieval, code generation, and exploration strategies. Third, adapting the architecture to additional EDA platforms, including Synopsys tools with Tcl scripting, to broaden the scope of execution-grounded EDA agents.

## References

[1] Anthropic. 2024. Model Context Protocol (MCP). https://modelcontextprotocol.io/.

[2] Jacob Austin, Augustus Odena, Maxwell Nye, Maarten Bosma, Henryk Michalewski, David Dohan, Ellen Jiang, Carrie Cai, Michael Terry, Quoc Le, et al. 2021. Program Synthesis with Large Language Models. *arXiv preprint arXiv:2108.07732* (2021).

[3] Jason Blocklove, Siddharth Garg, Ramesh Karri, and Hammond Pearce. 2024. Can EDA Tool Feedback Improve Verilog Generation by LLMs? *arXiv preprint arXiv:2411.11856* (2024).

[4] Jianlv Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. 2024. M3-Embedding: Multi-Linguality, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation. arXiv:2402.03216 [cs.CL] https://arxiv.org/abs/2402.03216

[5] Lei Chen, Yiqi Chen, Zhufei Chu, Wenji Fang, Tsung-Yi Ho, Yu Huang, Sadaf Khan, Min Li, Xingquan Li, Yun Liang, Yibo Lin, Jinwei Liu, Yi Liu, Guojie Luo, Zhengyuan Shi, Guangyu Sun, Dimitrios Tsaras, Runsheng Wang, Ziyi Wang, Xinming Wei,

Zhiyao Xie, Qiang Xu, Chenhao Xue, Evangeline F Y Young, Bei Yu, Minjie Yuan, Haoyi Zhang, Zuodong Zhang, Yuxiang Zhao, Hui-Ling Zhen, Ziyang Zheng, Binwu Zhu, Keren Zhu, and Sunan Zou. 2024. The Dawn of AI-Native EDA: Promises and Challenges of Large Circuit Models. *arXiv preprint arXiv:2403.07257* (2024). https://arxiv.org/abs/2403.07257

[6] Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, et al. 2021. Evaluating Large Language Models Trained on Code. *arXiv preprint arXiv:2107.03374* (2021).

[7] Cline Contributors. 2024. Cline: Autonomous AI Coding Assistant. https://github.com/cline/cline.

[8] Matthijs Douze, Alexandr Guzhva, Chengqi Deng, Jeff Johnson, Gergely Szilvasy, Pierre-Emmanuel Mazaré, Maria Lomeli, Lucas Hosseini, and Hervé Jégou. 2024. The Faiss library. (2024). arXiv:2401.08281 [cs.LG]

[9] Empyrean Technology. 2026. Empyrean PyAether: Full Custom Design Platform Ecosystem. https://empyrean.com.cn/solutions/index-128.html. Accessed: 2026-06-28.

[10] Xue Jiang, Yihong Dong, Zhi Jin, and Ge Li. 2024. SEED: Customize Large Language Models with Sample-Efficient Adaptation for Code Generation. *arXiv preprint arXiv:2403.00046* (2024).

[11] Leslie Pack Kaelbling, Michael L. Littman, and Anthony R. Cassandra. 1998. Planning and acting in partially observable stochastic domains. *Artificial Intelligence* 101, 1-2 (1998), 99–134.

[12] Rajesh Kumar et al. 2026. AgentForge: Execution-Grounded Multi-Agent LLM Framework for Autonomous Software Engineering. *arXiv preprint arXiv:2604.13120* (2026). https://arxiv.org/abs/2604.13120

[13] Qiufeng Li, Rongqian Chen, Quan Cheng, Chengxuan Wang, Sizhe Tang, Wuxi Li, Duo Ding, Chia-Tung Ho, Haoxing Ren, David Z. Pan, Tian Lan, and Weidong Cao. 2026. PDAGENT-BENCH: Characterizing, Grounding, and Architecting LLM Agents for VLSI Physical Design. *arXiv preprint arXiv:2606.17253* (2026).

[14] Bingyang Liu, Haoyi Zhang, Xiaohan Gao, Zichen Kong, Xiyuan Tang, Yibo Lin, Runsheng Wang, and Ru Huang. 2025. LayoutCopilot: An LLM-Powered Multiagent Collaborative Framework for Interactive Analog Layout Design. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)* 44, 8 (2025), 3126–3139. doi:10.1109/TCAD.2025.3529805

[15] Shang Liu, Wenji Fang, Yao Lu, Qijun Zhang, Hongce Zhang, and Zhiyao Xie. 2024. RTLCoder: Outperforming GPT-3.5 in Design RTL Generation with Our Open-Source Dataset and Lightweight Solution. In *2024 IEEE/ACM International Conference on Computer Aided Design (ICCAD)*. IEEE, New York, NY, USA. Also available at arXiv:2312.08617. https://arxiv.org/abs/2312.08617

[16] Yiyi Lu, Hoi Ian Au, Junyao Zhang, Jingyu Pan, Yiting Wang, Ang Li, Jianyi Zhang, and Yiran Chen. 2025. AutoEDA: Enabling EDA Flow Automation through Microservice-Based LLM Agents. *arXiv preprint arXiv:2508.01012* (2025). Uses MCP protocol for EDA automation. https://arxiv.org/abs/2508.01012

[17] OpenAI. 2023. *GPT-4 Technical Report*. Technical Report arXiv:2303.08774. OpenAI.

[18] Yuan Pu, Zhuolun He, Tairu Qiu, Haoyuan Wu, and Bei Yu. 2024. Customized Retrieval Augmented Generation and Benchmarking for EDA Tool Documentation QA. In *Proceedings of the IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*. IEEE, New York, NY, USA. arXiv:2407.15353 [cs.CL]

[19] Yibo Rui, Yuanhang Li, Rui Wang, Ruiqi Chen, Yanxiang Zhu, Zhixiong Di, Xi Wang, and Ming Ling. 2025. ChaTCL: LLM-Based Multi-Agent RAG Framework for TCL Script Generation. In *Proceedings of the International Symposium of Electronics Design Automation (ISEDA)*. IEEE, New York, NY, USA, 736–742.

[20] Cadence Design Systems. 2024. *Virtuoso Studio SKILL Language Reference*. https://www.cadence.com

[21] Shailja Thakur, Baleegh Ahmad, Hammond A Pearce, Benjamin Tan, Brendan Dolan-Gavitt, Ramesh Karri, and Siddharth Garg. 2023. VeriGen: A Large Language Model for Verilog Code Generation. *ACM Transactions on Design Automation of Electronic Systems* 29, 3 (2023), 1–31.

[22] Sebastian Thrun, Wolfram Burgard, and Dieter Fox. 2005. *Probabilistic Robotics*. MIT Press.

[23] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Lukasz Kaiser, and Illia Polosukhin. 2017. Attention is All You Need. *Advances in Neural Information Processing Systems* 30 (2017), 5998–6008.

[24] Ning Xu, Zhaoyang Zhang, Senlin Shu, Qi Lei, Jiaqi Lv, Wensuo Wang, Tianhao Zhao, Chao Zhang, Zhaoliang Yang, Xiangyu Li, Zhaorui Su, Jingshan Li, and Xin Geng. 2026. iScript: A Domain-Adapted Large Language Model and Benchmark for Physical Design Tcl Script Generation. *arXiv preprint arXiv:2603.04476* (2026).

[25] An Yang et al. 2025. Qwen3 Technical Report. arXiv:2505.09388 [cs.CL] https://arxiv.org/abs/2505.09388